

From Data-Driven to Purpose-Driven Machine Learning in Healthcare

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MSc Methodology & Statistics
(Applied Data Science Track)

BSc Psychology

Natural language processing, mental
disorder prediction from social
media data, and explainable
machine learning



Philosophy

- I am **optimistic** about the potential of machine learning to benefit patient care
- I recognize that there is a plethora of **sociotechnical challenges** that need to be overcome before we reach the more automated future
- **Computational methods** can help align AI systems with socially desirable outcomes, however, these methods have to be **grounded in sociotechnical realities** of operational settings
- In my work, I seek to combine **machine learning** and **systems thinking** to propose methods that have real-world value & impact

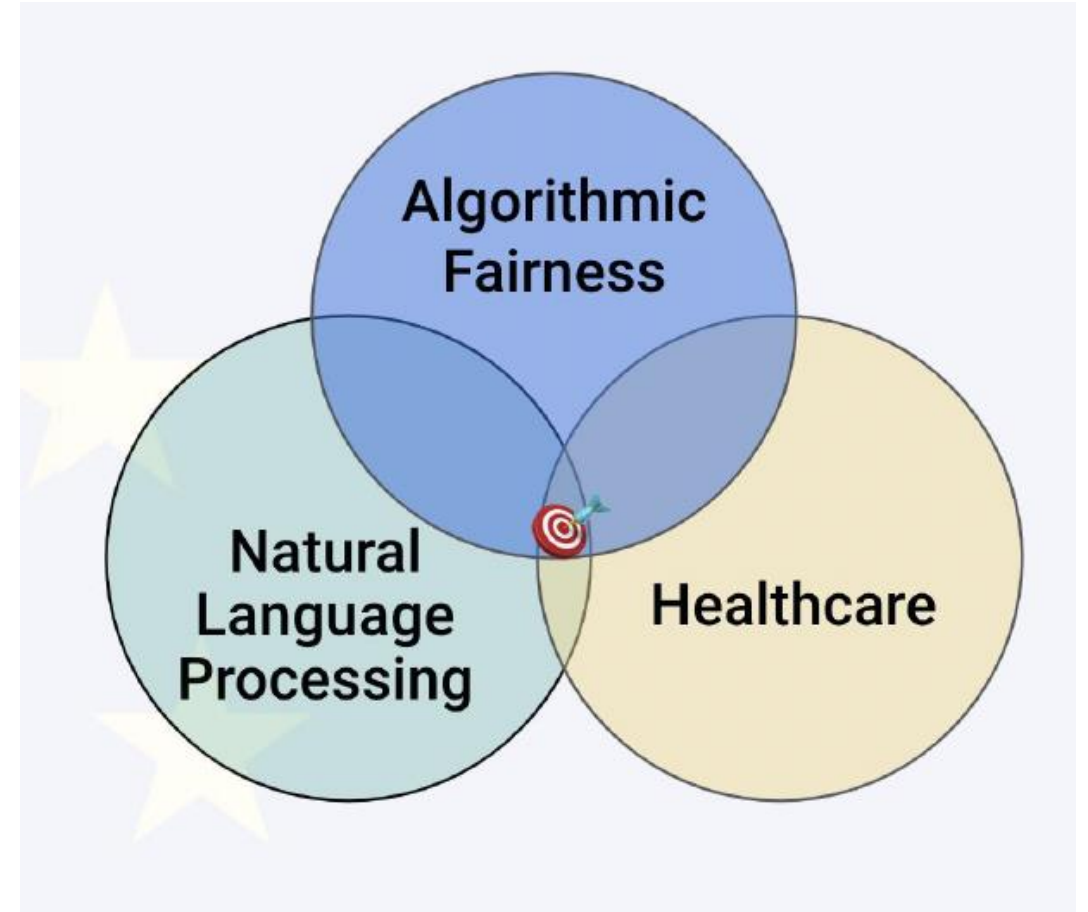
Sample of My Projects

- Algorithmic Fairness in Natural Language Processing
- Values Permeating Data Generation: Case of EEG Reports
- Systems Thinking for Data-Analytic Automation of Patient Care

Algorithmic Fairness in Clinical Natural Language Processing: Challenges & Opportunities

Algorithmic Fairness in Clinical Natural Language Processing: Challenges & Opportunities

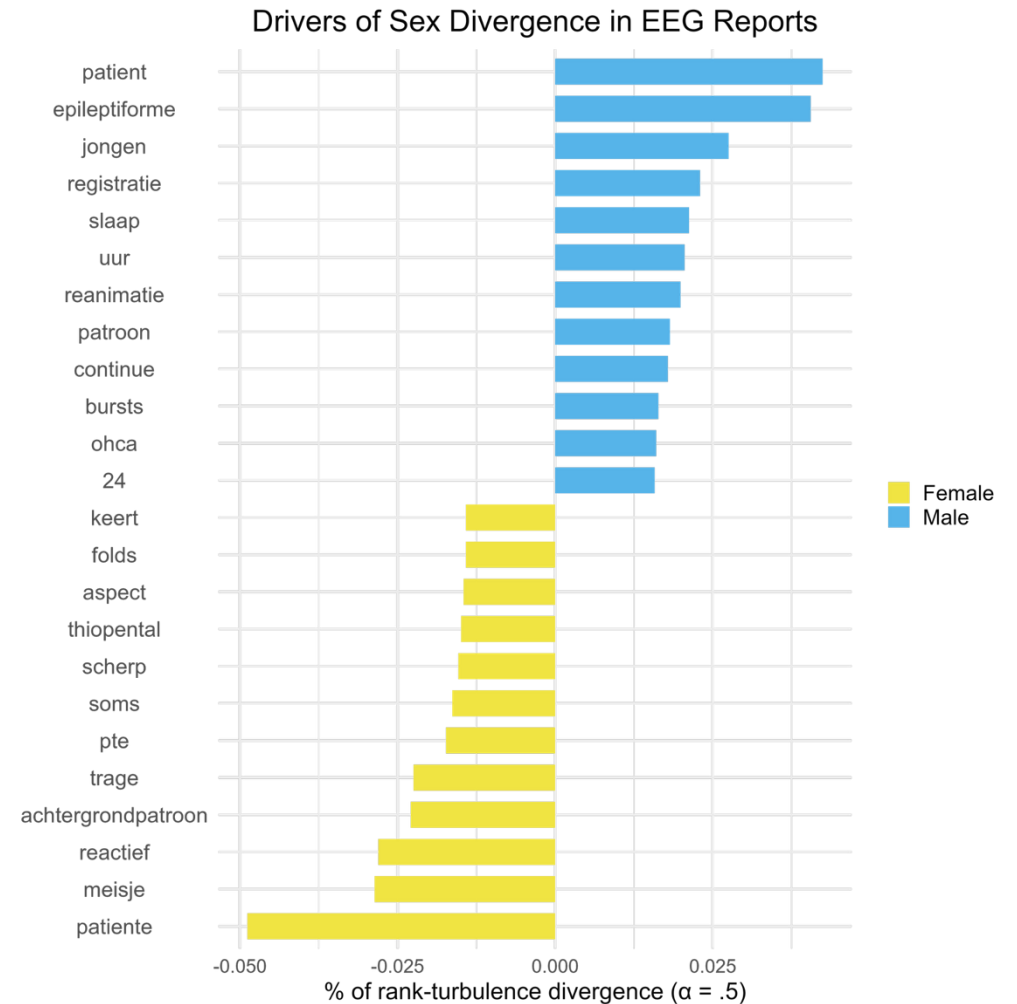
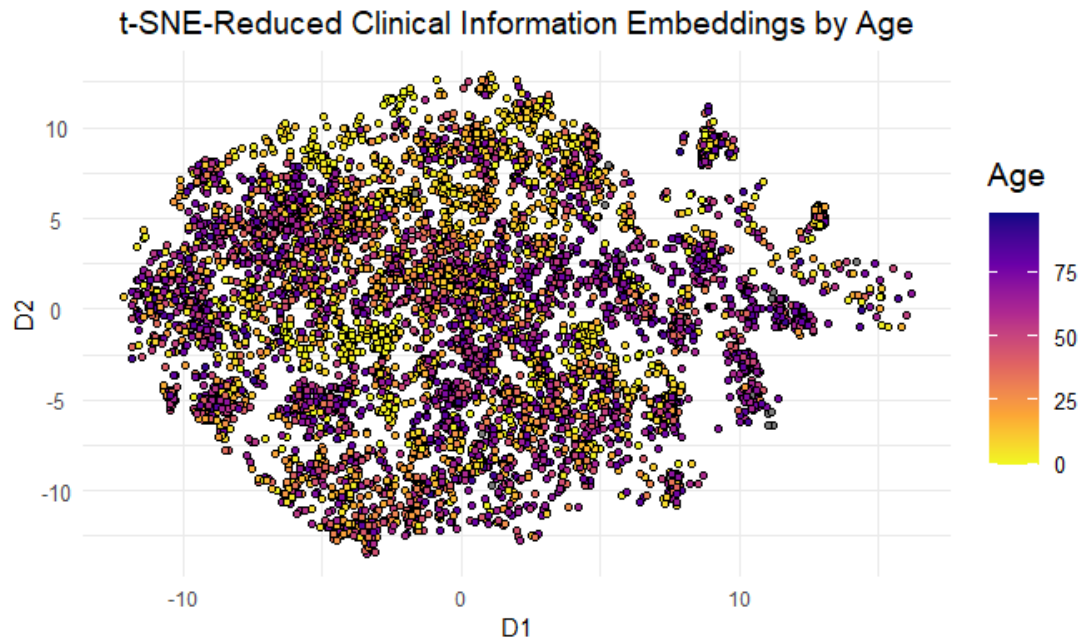
- Scoping review of literature aimed at identifying research gaps
- Identified gaps were associated with protected groups, method selection, data sharing and privacy, and generalizability
- Work presented & published in proceedings of the Third European Workshop on Algorithmic Fairness (EWAF'24)



Values Permeating Data Generation: Case of EEG Reports

Values Permeating Data Generation: Case of EEG Reports

- Internship at St. Antonius Hospital neurology department in Utrecht
- Studied EEG reports for group differences using natural language processing methods

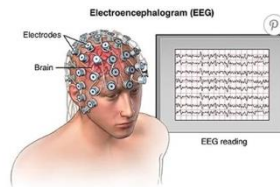


Values Permeating Data Generation: Case of EEG Reports

- We also mapped the diagnostic workflows at the neurology department
- EEG data come from a mixture of data generating distributions

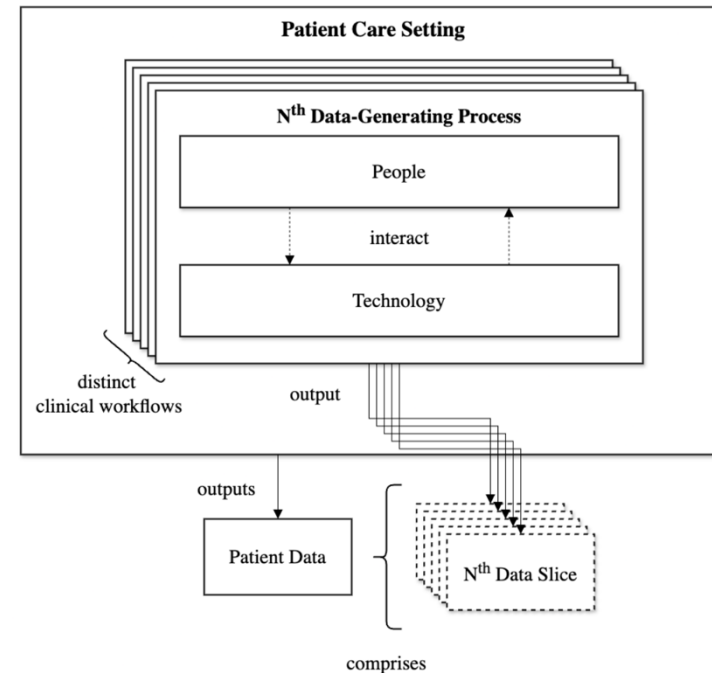
Diagnostic Questions

- Epilepsy diagnosis after first seizure-like event
- Neurological prognosis after cardiac arrest at ICU
- Cognitive disorders
- Dementia
- Encephalopathy
- Delirium



EEG Data

Figure 5: Most real-world patient datasets arise from a mixture of distinct data-generating processes.



Real-world patient data are often reflect a mixture of several data-generating processes. Linking individual data tances to those clinical processes which had generated them is not always straightforward. Indicators of clinical processes may be implicit, latent, and not clearly understood by all stakeholders which use patient data.

Values Permeating Data Generation: Case of EEG Reports

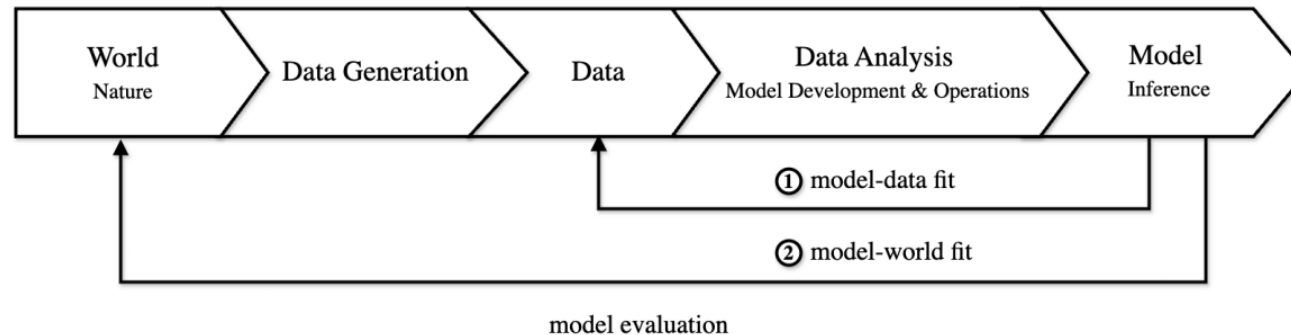
- When there is uncertainty in the presentation of an EEG and medical history, the clinical neurophysiologist has to decide whether to diagnose abnormalities (yes/no), or refer the patient for another EEG
- The clinician I worked with opts for labeling uncertain cases as 'no'. This strategy will reduce (false positive) costs, but it will also increase the false negative rate / patient underdiagnosis
- Following an uncertain EEG measurement, some individuals do not show up for the second EEG measurement
- The clinician I spoke to attributes that to the fact that diagnosed individuals lose the right to drive in the Netherlands
- This incentive leads to an underestimation of epilepsy in the population

Systems Thinking for Data-Analytic Automation of Patient Care

Systems Thinking for Data-Analytic Automation of Patient Care

- Understanding the utility of existing patient datasets requires looking in two directions: upstream towards data generation, and downstream towards machine learning objectives
- This is the first work to combine machine learning theory and systems theory in automation contexts
- We explored the functional and quality requirements of patient data as AI automation system components

Figure 1: Model development begins with nature that generates data. In data-analytic modeling, a sample of data, produced by a real-world data-generating process, is used develop a model. The model is a mathematical representation of a part of the joint distribution produced by the data-generating mechanism that is of interest, and can be evaluated by its fit to (1) the data, and (2) the world.



Systems Thinking for Data-Analytic Automation of Patient Care

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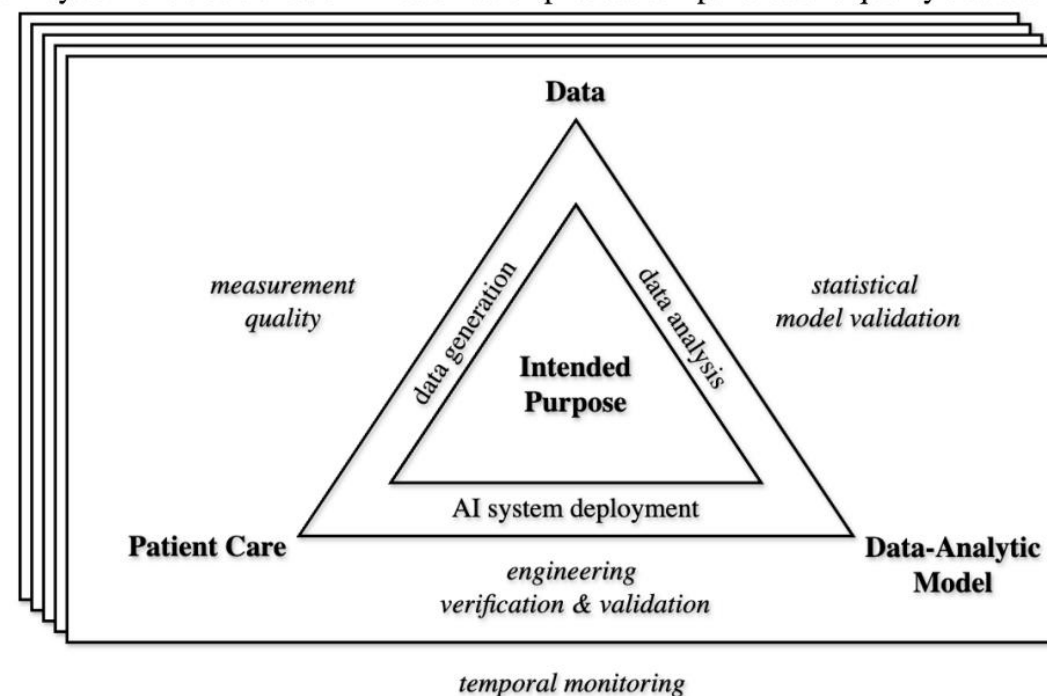
[Submitted on 16 Jun 2025 (v1), last revised 18 Jun 2025 (this version, v2)]

From Data-Driven to Purpose-Driven Artificial Intelligence: Systems Thinking for Data-Analytic Automation of Patient Care

Daniel Anadria, Roel Dobbe, Anastasia Giachanou, Ruurd Kuiper, Richard Bartels, Wouter van Amsterdam, Íñigo Martínez de Rituerto de Troya, Carmen Zürcher, Daniel Oberski

In this work, we reflect on the data-driven modeling paradigm that is gaining ground in AI-driven automation of patient care. We argue that the repurposing of existing real-world patient datasets for machine learning may not always represent an optimal approach to model development as it could lead to undesirable outcomes in patient care. We reflect on the history of data analysis to explain how the data-driven paradigm rose to popularity, and we envision ways in which systems thinking and clinical domain theory could complement the existing model development approaches in reaching human-centric outcomes. We call for a purpose-driven machine learning paradigm that is grounded in clinical theory and the sociotechnical realities of real-world operational contexts. We argue that understanding the utility of existing patient datasets requires looking in two directions: upstream towards the data generation, and downstream towards the automation objectives. This purpose-driven perspective to AI system development opens up new methodological opportunities and holds promise for AI automation of patient care.

Figure 5: A system-theoretic view of AI automation process components and quality evaluation procedures.



Articles

Anadria, D., Dobbe, R.I.J., Giachanou A., Kuiper R.J.A., Bartels, R.T. Martínez de Rituerto de Troya, Í., Zürcher, C., Oberski, D.L. (2025) From Data-Driven to Purpose-Driven Artificial Intelligence: Systems Thinking for Data-Analytic Automation of Patient Care. arXiv preprint.

<https://arxiv.org/abs/2506.13584>

Anadria, D. Giachanou, A. Kernahan, J.A., Dobbe, R.I.J., and Oberski, D.L. (2025) Algorithmic Fairness in Clinical Natural Language Processing: Challenges and Opportunities. In Proceedings of the 3rd European Workshop on Algorithmic Fairness, volume 3908, https://ceur-ws.org/Vol-3908/paper_50.pdf

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